

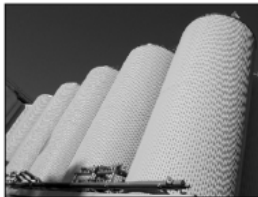
Detecting Simple Curves

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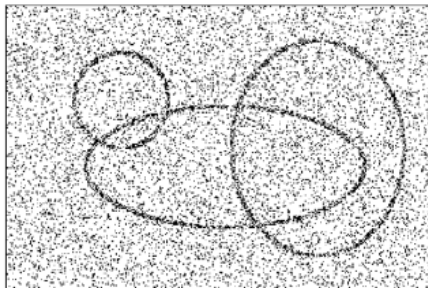
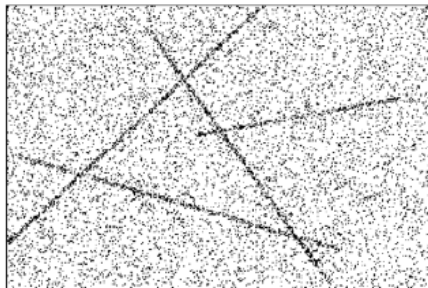
Detecting Lines and Simple Curves

- Many man-made objects exhibit simple geometric forms: lines, circles, ellipses



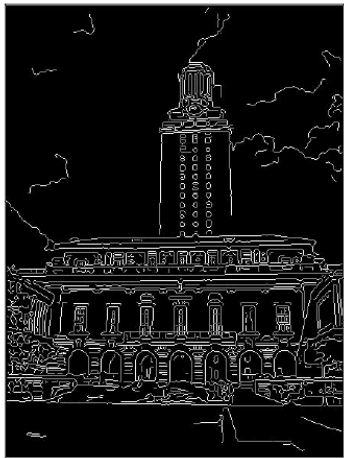
Hough Transform

- Hough transform: Find any shape that can be defined parametrically within a distribution of points (Paul Hough)
- Example: lines, circles, ellipses.
- Used to find line segments in edge maps



Why isn't displaying results of edge detection adequate

Difficulty of line fitting



- Extra edge points (clutter), multiple models:
 - which points go with which line, if any?
- Only some parts of each line detected, and some parts are missing:
 - how to find a line that bridges missing evidence?
- Noise in measured edge points, orientations:
 - how to detect true underlying parameters?

Hough Transform

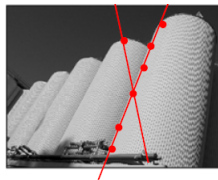
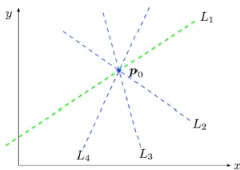
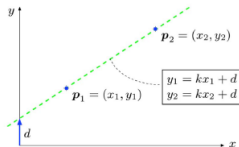
- Equation of a line

$$y = kx + d$$

- If points p_1 and p_2 on same line, same k and d

- Hough transform:

- start with edge point (x, y)
- Find (slope k , intercept d) that passes through the most edge points



Hough Transform

- Consider the point $(x,y) = (1,1)$
- Many lines of different slope k and intercept d can pass through $(1,1)$
- Thus we can rewrite equation of line through $(1,1)$ as $1 = k \cdot 1 + d(\text{image}) \rightarrow d = -k + 1(\text{parameter space})$
- Set of all lines passing through $(1,1)$ (left) can be shown as line in parameter space (right)

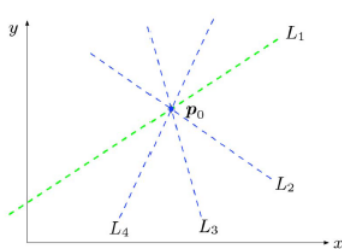
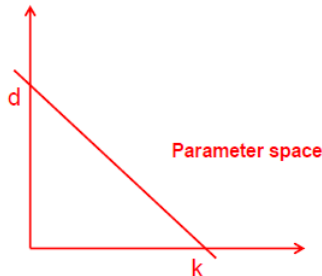


Image space



Hough Transform: Example

- Suppose image has 5 points: $(1,0)$, $(1,1)$, $(2,1)$, $(4,1)$ and $(3,2)$
- Each of these points corresponds to the following lines that can be plotted

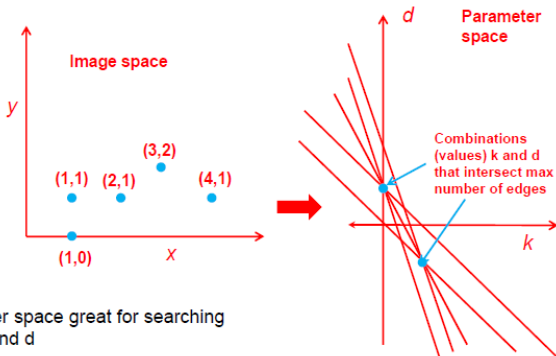
$$(1,0) \rightarrow d = -k$$

$$(1,1) \rightarrow d = -k + 1$$

$$(2,1) \rightarrow d = -2k + 1$$

$$(4,1) \rightarrow d = -4k + 1$$

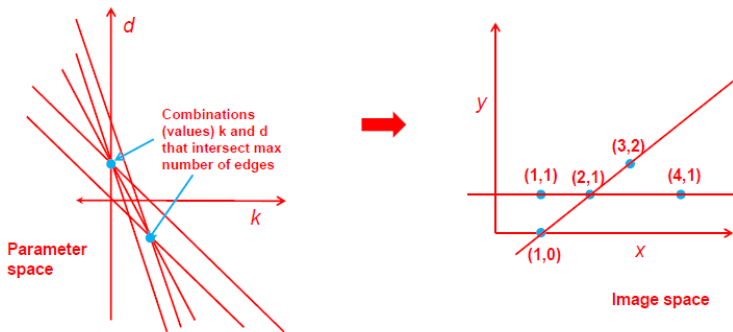
$$(3,2) \rightarrow d = -3k + 2$$



Note: Lines in parameter space great for searching for optimal values of k and d

Hough Transform: Example

- After finding optimal values of k and d , we can draw them as lines in image space
- In example: $(k, d) = (0, 1)$ and $(1, -1)$ each intersect 3 lines
- Converting to image space:
 $(0, 1) \rightarrow y = 1, (1, -1) \rightarrow y = x - 1$



Hough Transform: Image vs Parameter Space

- If N lines intersect at position (k', d') in **parameter space**, then N image points lie on the corresponding line $y = kx + d$ in **image space**

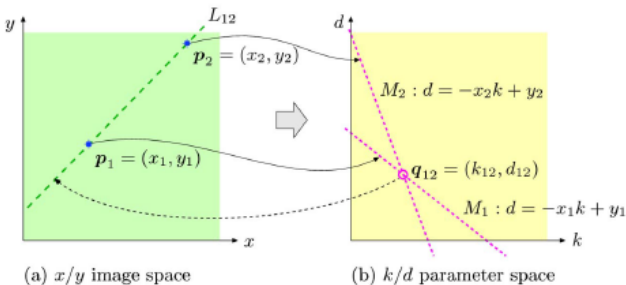
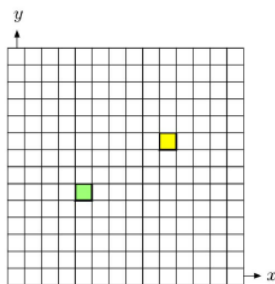


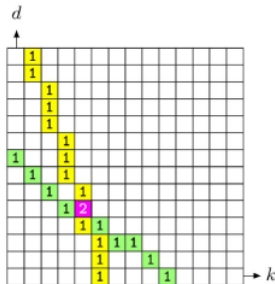
Image Space (x, y)		Parameter Space (k, d)	
Point	$p_i = (x_i, y_i)$	$M_i : d = -x_i k + y_i$	Line
Line	$L_j : y = k_j x + d_j$	$q_j = (k_j, d_j)$	Point

Accumulator Array

- Accumulator array: discrete representation of parameter space as 2D array
- Given a point in image, increment all points on it's corresponding line (draw line) in parameter space
- A line in image will be intersection of multiple lines in parameter space.



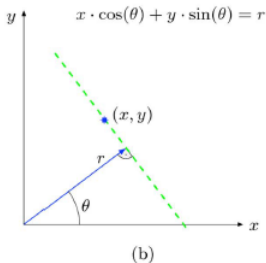
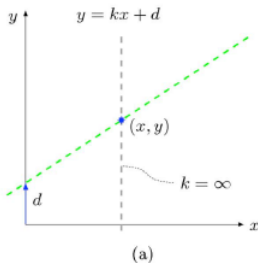
(a) Image Space



(b) Accumulator Array

Hough Transform

- A Problem: The form of line cannot express vertical line (slope $k = \text{infinity}$)
- Alternate line representation in terms of r, θ :
 - r , perpendicular distance of line from origin
 - θ , angle of line's perpendicular to x axis
 - If we allow negative r , then $-90^\circ < \theta < 90^\circ$



Hough Transform

- Point (p, q) where perpendicular to line meets line = $(r \cos \theta, r \sin \theta)$
- Consider (x, y) , another point on line, gradient of line is

$$\frac{y - q}{x - p} = \frac{y - r \sin \theta}{x - r \cos \theta}$$

- Gradient of perpendicular is $\tan \theta$, so gradient of line is $-\frac{1}{\tan \theta} = -\frac{\cos \theta}{\sin \theta}$
- Putting these two expressions together

$$\frac{y - r \sin \theta}{x - r \cos \theta} = -\frac{\cos \theta}{\sin \theta}$$

- If we multiply these fractions, we obtain

$$y \sin \theta - r \sin^2 \theta = -x \cos \theta + r \cos^2 \theta$$

rewritten as

$$y \sin \theta + x \cos \theta = r \text{ (called Hessian Normal Form)}$$

Implementing Hough Transform using Hessian Normal Form of line

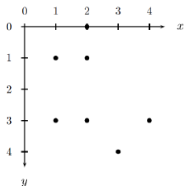
- Each point (x,y) that falls on line with (r, θ) must satisfy equation below

$$y \sin \theta + x \cos \theta = r$$

- To implement Hough transform,
 - Choose discrete set of θ values
 - For instance, for image shown consider θ values

$$-45^0, 0^0, 45^0, 90^0$$

- For each pixel (x,y) in image, compute r
- Put results in a table



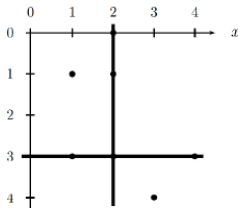
(x, y)	-45°	0°	45°	90°
(2, 0)	1.4	2	1.4	0
(1, 1)	0	1	1.4	1
(2, 1)	0.7	2	2.1	1
(1, 3)	-1.4	1	2.8	3
(2, 3)	-0.7	2	3.5	3
(4, 3)	0.7	4	4.9	3
(3, 4)	-0.7	3	4.9	4

Implementing Hough Transform using Hessian Normal Form of line

- Accumulator array (below): contains number of times each value of (r, θ) appears in table (above)

	-1.4	-0.7	0	0.7	1	1.4	2	2.1	2.8	3	3.5	4	4.9
-45°	1	2	1	2		1							
0°					2		3			1		1	
45°						2		1	1		1		2
90°			1		2					3		2	

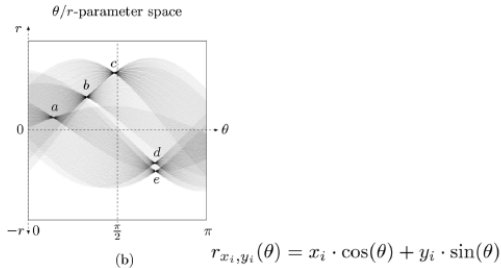
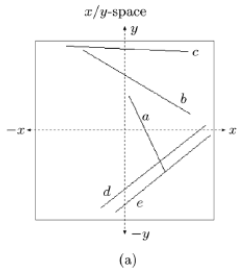
- Two largest values occur at
 $(r, \theta) = (2, 0^\circ)$ and $(r, \theta) = (3, 90^\circ)$
- The corresponding lines are: $x = 2$ or $y = 3$



Visualizing Image Space to Parameter Space Representation

- Image Space and parameter space using HNF representation for $0 \leq \theta < \pi$
- If image center is reference, then range of r limited to half of diagonal. E.g. for image of width M and height N

$$-r_{\max} \leq r_{x,y}(\theta) \leq r_{\max}, \quad \text{where} \quad r_{\max} = \frac{1}{2} \sqrt{M^2 + N^2}$$



- Digital Image Processing (CS/ECE 545), Computer Science Dept. Worcester Polytechnic Institute (WPI)